

In the name of god

CS-Pre Lab No. 6

UART and SPI Communication

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1. UART Communication

- Draw the timing diagram of a UART communication sequence (without RTS and CTS signals). Label start bits, data bits, stop bits, and any parity bits.
- Decode the timing diagram you created and explain the meaning of the various components (e.g., stop bits and parity bits).

Configuration

We consider the following UART settings:

- Baud Rate: 9600 bps
- Data Bits: 8
- Parity Bit: Even
- Stop Bits: 1
- Transmitted Character: 'A' (ASCII 0x41 = 01000001)

UART sends data LSB first, so the data bits transmitted are: 10000010.

Q1: Timing Diagram and UART Transmission of Character 'A'

- Character: 'A' (ASCII 0x41)
- Binary: 01000001
- LSB first : 10000010
- Baud rate: 9600 bps

- Frame: Idle- Start- Data [8]- Parity- Stop

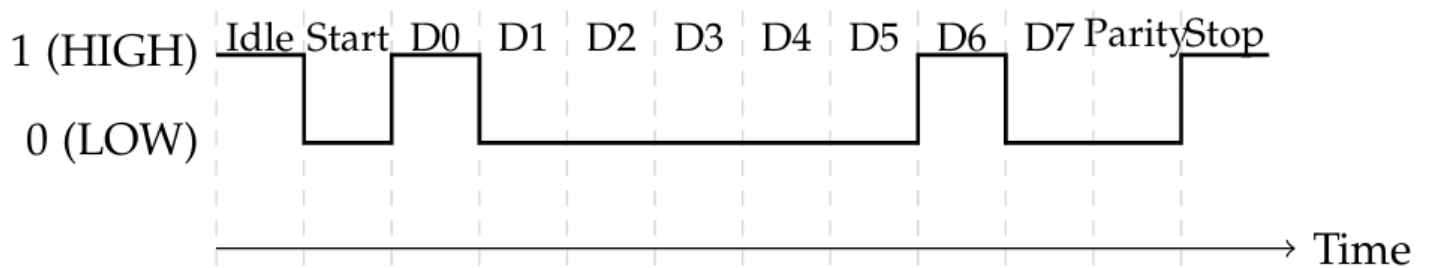


Figure1

Q2: Frame Structure Breakdown

- Idle (High): Line is high before transmission begins.
- Start Bit (Low): Signals the start of transmission.
- Data Bits: Sent LSB first. For 'A' = 01000001, this is 10000010.
- Parity Bit (Even): Data has two 1s, so parity bit is 0 to keep total even.
- Stop Bit (High): Marks end of frame; line remains high afterwards.

Notes

- Each bit lasts for approximately $104.17\mu\text{s}$ at 9600 bps.
- The receiver samples each bit at the middle of the bit time.
- Parity bit allows detection of single-bit errors.
- UART is asynchronous and requires both transmitter and receiver to agree on baud rate.

2. SPI Communication

- Name three sensors that utilize SPI and briefly explain their functionalities. You may look at the appendix-A part of the c-sdk reference document for this.
- Draw a timing diagram for SPI communication. You are free to select a CPOL and CPHA scheme of your liking.
- Label the roles of each SPI signal pin (SCLK, MOSI, MISO, SS/CS) on the timing diagram.

Q1: SPI-Based Sensors and Their Functionalities

MPU9250 9-axis Motion Sensor:

The MPU9250 integrates a 3-axis gyroscope, a 3 axis accelerometer, and a 3-axis magnetometer. It is widely used in applications such as drones, augmented reality, and robotics. SPI allows the fast transmission of sensor data (up to several MHz), reducing latency and improving system response.

MAX6675 K-type Thermocouple Sensor:

This sensor measures temperatures up to 1024°C and outputs digital values directly over the SPI interface. Its built-in ADC eliminates the need for analog signal processing in the microcontroller.

MFRC522 RFID Reader:

Used for contactless communication with RFID tags, this module uses SPI for rapid data transfer. It is commonly found in

secure access systems, smart cards, and embedded NFC solutions.

Q2: SPI Timing Diagram

SPI Mode: CPOL = 0, CPHA = 0

In SPI Mode 0:

- CPOL=0: Clock idle state is LOW.
- CPHA = 0: Data is sampled on the rising edge of SCLK.

The following diagram shows an 8-bit full duplex SPI transaction:

- MOSI (Master Out, Slave in) transmits: 11001010
- MISO (Slave Out, Master in) returns: 10110100

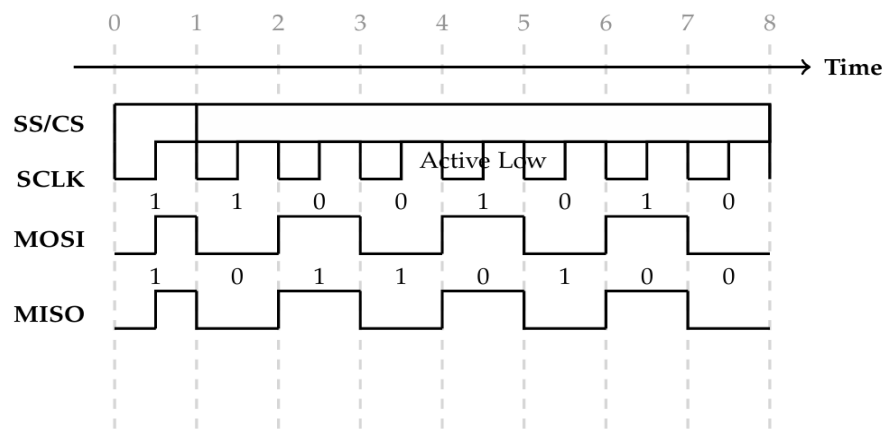


Figure2: SPI Timing Diagram (CPOL = 0, CPHA = 0). Data is sampled on the rising edge of the clock (SCLK). Both MOSI and MISO operate simultaneously in full-duplex mode. SS/CS is active LOW throughout the transmission.

Notes:

- Each clock cycle (2C) represents one-bit transfer.
- Transitions occur on falling edge, sampling occurs on rising edge (CPHA = 0).

Q3: SPI Signal Pin Roles

- SCLK (Serial Clock): Clock signal generated by the master to synchronize data transmission.
- MOSI (Master Out, Slave in): Transfers data from master to slave.
- MISO (Master in, Slave Out): Transfers data from slave to master.
- SS/CS (Slave Select / Chip Select): Active LOW signal used to select the target slave device.